

LESSON 9.8

EQUIVALENT EXPRESSIONS USING PROPERTIES OF OPERATIONS



LESSON INTRODUCTION

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.

LEARNING TARGETS

- I can use properties of operations to write equivalent forms of an algebraic expression.
- I can determine whether two linear expressions are equivalent.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

WORKING TOWARD

MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

ESSENTIAL QUESTIONS

- How can you prove linear expressions are equivalent using properties of operations?

LESSON NARRATIVE

Students extend the use of properties from numeric to algebraic expressions. Students begin by refining their working definitions of commutative, associative, and distributive properties in the context of numeric expressions before exploring what those properties look like through the lens of algebraic expressions. This lesson includes more spiraled practice on fluency with fractions.

COMPONENT SUMMARY

9.8.1 WARM-UP (~5 MIN.)

Fact or Fiction? equivalent expressions

9.8.3 GUIDED INSTRUCTION (10 MIN.)

Extending properties from numeric to algebraic expressions

9.8.5 YOUR TURN! (10 MIN.)

Students choose properties to rewrite equivalent algebraic expressions

9.8.2 REFRESHER (10 MIN.)

Reviewing associative, commutative, and distributive properties of numeric expressions to apply to algebraic expressions

9.8.4 GUIDED PRACTICE (10 MIN.)

Write equivalent algebraic expressions using the properties of operations

9.8.6 WRAP-UP (~5 MIN.)

Determine if two linear expressions are equivalent

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Associative property
- Distributive property
- Commutative property

MATERIALS

- (Optional) Note cards
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- Consider a Gallery Walk with chart paper for 9.8.2 Refresher.

TEACHER TIPS

COMMON MISCONCEPTIONS

- When rewriting expressions using the distributive property, students may forget to distribute to all terms.
- Students may add instead of multiply coefficient and substituted values, like $3x \rightarrow 32 \rightarrow 5$, or fail to use parentheses and interpret this as 32.

THINGS TO CONSIDER

- Similarly, for variable expressions, encourage students to substitute values into the expressions and evaluate each expression to verify that the expressions are equivalent.
- Colors may be helpful when substituting values in for the variables.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 9.8.4 Guided Practice, students create equivalent expressions using properties of operations. Prior to independently recording the properties, ask students to engage in a Think-Pair-Share to swap their expressions with a partner and determine the properties their partners used.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

After reviewing the associative, commutative, and distributive properties, students have multiple opportunities to apply, extend, and practice the properties. During partner work, encourage students to reflect on their own choice of properties to rewrite the expressions and compare that to those of their partner.

DIFFERENTIATE INSTRUCTION

9.8.5 Your Turn! offers student choice and multiple pathways as students apply any of the properties to different expressions. Consider a Four Corners activity in which each property is assigned a corner. Students must find a classmate from a different corner, show their work, and verify the work of another. Alternatively, students can hold up a number from one to three, with each number representing which property they applied. Students then must pair up with a classmate of the same number to verify their answers before finding a new partner with a different number and comparing their work.

9.8.1 WARM-UP: FACT OR FICTION?

ENRICHMENT

Consider prompting questions to compare methods of evaluation and properties.

- Do you have to evaluate (or simplify) to determine if both sides of the equation are equivalent?
- Are the sides of each equation equal or equivalent?



9.8.1 Warm-Up: Fact or Fiction?

1. Determine whether each equation below is fact (true) or fiction (false).

Equation	Fact or Fiction
$(30 + 8) \cdot 9 = 270 + 72$	☐ Fact ☐ Fiction
$4 + 5(9 - 15) = 9(9 - 15)$	☐ Fact ☐ Fiction
$2 \cdot (7 + 8) = (2 \cdot 7) + 8$	☐ Fact ☐ Fiction
$(5 + 4) - 10 = 5 + (4 - 10)$	☐ Fact ☐ Fiction



9.8.2 Refresher: Rewriting Expressions Using Properties of Operations

- Fill in the missing blanks for each property.
 - Using the **commutative property of addition**, $3 + 12$ can be rewritten as $12 + 3$. Both expressions have a sum of 15.
 - Using the **commutative property of addition**, $4x + 5$ can be rewritten as $5 + 4x$. Both expressions will have the same sum regardless of the value of x .
 - Using the **commutative property of multiplication**, $(4)(-2)$ can be rewritten as $(-2)(4)$. Both expressions have a product of -8.
 - Using the **commutative property of multiplication**, $(-3y)(0.5)$ can be rewritten as $(0.5)(-3y)$. Both expressions will have the same product regardless of the value of y .
 - Using the **associative property of addition**, $-5 + (6 + 1)$ can be rewritten as $(-5 + 6) + 1$. Both sums are equal to 2.
 - Using the **associative property of addition**, $9x + (-2x + 5)$ can be rewritten as $(9x + -2x) + 5$. Both expressions will have the same sum regardless of the value of x .
 - Using the **associative property of multiplication**, $2 \cdot (7 \cdot 3)$ can be rewritten as $(2 \cdot 7) \cdot 3$. Both products are equal to 42.
 - Using the **associative property of multiplication**, $(-6)(4y)$ can be rewritten as $(-6 \cdot 4)y$. Both expressions will have the same product regardless of the value of y .

- Using the **distributive property of multiplication over addition**, $5(6 + 2)$ can be rewritten as $(5 \cdot 6) + (5 \cdot 2)$. Both expressions have a value of 40.
 - Using the **distributive property of multiplication over addition**, $7(3y + 8)$ can be rewritten as $(7 \cdot 3y) + (7 \cdot 8)$ or $21y + 56$. Both expressions will have the same product regardless of the value of y .
- Discuss with your partner why the associative property and the commutative property do not hold true for subtraction and division. Give an example of each to show mathematically that these properties do not apply to subtraction and division.

The associative property does not hold true for subtraction. For example: $(10 - 5) - 2 = 3$ is not equal to $10 - (5 - 2) = 7$.

The associative property does not hold true for division. For example: $(50 \div 5) \div 2 = 5$ is not equal to $50 \div (5 \div 2) = 20$.

The commutative property does not hold true for subtraction. For example: $10 - 5 = 5$ is not equal to $5 - 10 = -5$.

The commutative property does not hold true for division. For example: $10 \div 5 = 2$ is not equal to $5 \div 10 = \frac{1}{2}$.
 - Jay gave the example $6 + -11$ is equal to $-11 + 6$ to show that subtraction is commutative. Explain to Jay why his example does not show that subtraction is commutative.

Jay's problem shows that the commutative property holds for addition.

9.8.2 REFRESHER: REWRITING EXPRESSIONS USING PROPERTIES OF OPERATIONS

DIFFERENTIATE INSTRUCTION

Data-Driven Instruction

Allow data and informal observations from previous lessons in this unit to drive decision-making.

- Small Groups: Utilize a teacher-led small group if you have a handful of students who need additional support.
- Gallery Walk: Create three sets of all three properties (nine total so each poster only has three to four students at one time) on chart paper. Students must visit each property and write at least one example of that property. After they write one example for each property, they then must find a peer's examples and verify by writing a check mark if correct.
- Collaboration: Consider grouping students strategically based on the 9.7.7 Wrap-Up from the previous lesson if misconceptions are common or have trends.
- Independent: Utilize independent practice if students need time meaningfully engaging in producing examples. These are best utilized in context, so prioritize question 2 more than question 1.

DIFFERENTIATE INSTRUCTION

Consider providing sentence starters for students who may struggle with written explanations. Alternatively, allow students to explain their responses orally to provide an additional entry point.

9.8.3 GUIDED INSTRUCTION: EQUIVALENT EXPRESSIONS USING PROPERTIES OF OPERATIONS

ENRICHMENT

How can you use properties of operations to determine whether algebraic expressions are equivalent?

QUESTIONING

For students who may finish early, encourage them to rewrite the expression in question 2 in its simplest form and explain how they know it is in simplest form.



9.8.3 Guided Instruction: Equivalent Expressions Using Properties of Operations

Algebraic expressions are considered equivalent if when evaluating the expression by substituting values in for variables, the resulting values are the same.

Algebraic expressions are also considered equivalent when one expression is rewritten using one or more of the properties of operations.

1. Complete each statement.

- a. The
☐ associative
☐ commutative
☒ distributive
 property can be used to state that $5(x + 2) - 3x$ is equivalent to $5x + 10 - 3x$.
- b. The
☐ associative
☒ commutative
☐ distributive
 property can be used to state that $\frac{2}{3}x + (\frac{3}{8} + \frac{5}{6}x)$ is equivalent to $\frac{2}{3}x + (\frac{5}{6}x + \frac{3}{8})$.

2. Which properties can be used to show that $2(x - 4) + 5(3 + 6x)$ is equivalent to $2x + 30x - 8 + 15$?

The distributive property of multiplication over addition and over subtraction were used to obtain $2x - 8 + 15 + 30x$. Then, the commutative property was used to rearrange the terms and obtain $2x + 30x - 8 + 15$.

9.8.4 GUIDED PRACTICE: REWRITING EXPRESSIONS USING PROPERTIES OF OPERATIONS

TEACHER MOVES

Think-Pair-Share

Encourage mathematical discourse by implementing a Think-Pair-Share to position students as property experts in creating, verifying, and justifying responses.

QUESTIONING

Challenge students to write an equivalent expression that uses all three properties of operations (the associative, commutative, and distributive properties).



9.8.4 Guided Practice: Rewriting Expressions Using Properties of Operations

1. Write an equivalent expression using a property of operations. State the property used.

Answers will vary.

Expression	Equivalent Expression	Property Used
$-6(2 - 3x) + 1 - \frac{x}{2}$	$= -12 + 18x + 1 - \frac{1}{2}x$ $= 18x - \frac{1}{2}x - 12 + 1$ $= 17\frac{1}{2}x - 11$	<i>Distributive & Commutative Property</i>
$12x + 10 + 4x - 17$	$= 16x - 7$	<i>Commutative Property</i>
$5x + (3x + 9) + 2$	$= 8x + 11$	<i>Associative Property</i>

2. Several expressions are given below. Determine which expressions are equivalent using properties of operations.

$35x + (7x + 56)$	$5(8x + 7)$	$35x + 56$
$(8x + 7) \cdot 5$	$7(5x + 8)$	$(35x + 7x) + 56$

Complete the table by writing the expressions that are equivalent based on the property used.

Equivalent Expressions		Property Used
$35x + (7x + 56)$	$(35x + 7x) + 56$	Associative Property
$5(8x + 7)$	$(8x + 7) \cdot 5$	Commutative Property
$35x + 56$	$7(5x + 8)$	Distributive Property

**9.8.5 Your Turn!**

Write an equivalent expression using the associative, commutative, or distributive property. State the property used.

Answers will vary.

- $10x + \frac{2}{5}(-30 + 15x)$
 $= 10x + -12 + 6x$
Distributive property of multiplication over addition
- $(18x + 15) + 4x - 17$
 $= (15 + 18x) + 4x - 17$
Commutative property of addition
- $(-\frac{1}{5}x + 6) \cdot 5$
 $= 5 \cdot (-\frac{1}{5}x + 6)$
Commutative property of multiplication
- $\frac{5}{4}n + \frac{7}{3} - \frac{11}{2}n - 12\frac{1}{3}$
 $= \frac{5}{4}n + -\frac{11}{2}n + \frac{7}{3} + -12\frac{1}{3}$
Commutative property of addition
- $-3(\frac{1}{2}x + 4) - \frac{3}{5}(-1 + 10x)$
 $= -\frac{3}{2}x - 12 - \frac{3}{5}(-1 + 10x)$
Distributive property of multiplication over addition

9.8.5 YOUR TURN!**TEACHER MOVES**

There is more than one correct equivalent expression. Ask students to share and justify how they used properties to write an equivalent expression and consider having students evaluate the examples they have come up with as a class to encourage analysis. Correct student responses could be used to create a Card Sort as a review activity before the unit assessment.

DIFFERENTIATE INSTRUCTION

Consider providing a specific property to use to rewrite each expression. This will encourage students to practice all three properties and may provide an entry point for students who may struggle getting started on their own.

**9.8.6 Wrap-Up: Ticket Out the Door**

Use properties of operations to determine if the expressions are equivalent.

- $\frac{4}{5}(x - 1) - \frac{1}{8}(2x + 1)$ and $\frac{11}{20}x - \frac{37}{40}$
The expressions are equivalent.

$$\begin{aligned} &\frac{4}{5}x - \frac{4}{5} - \frac{2}{8}x - \frac{1}{8} \\ &= \left(\frac{4}{5}x - \frac{2}{8}x\right) + \left(-\frac{4}{5} - \frac{1}{8}\right) \\ &= \left(\frac{11}{20}x\right) + \left(-\frac{37}{40}\right) \\ &= \frac{11}{20}x - \frac{37}{40} \end{aligned}$$
- $1\frac{3}{10}x + \frac{1}{5}x - \frac{1}{10} - x - \frac{1}{10}$ and $1\frac{1}{10} - \frac{4}{5}x$
The expressions are not equivalent.

$$\begin{aligned} &\left(1\frac{3}{10}x + \frac{1}{5}x - x\right) + \left(-\frac{1}{10} - \frac{1}{10}\right) \\ &= \left(\frac{3}{10}x + \frac{2}{10}x\right) + \left(-\frac{2}{10}\right) \\ &= \left(\frac{5}{10}x\right) + \left(-\frac{2}{10}\right) \\ &= \frac{1}{2}x - \frac{1}{5} \end{aligned}$$

9.8.6 WRAP-UP: TICKET OUT THE DOOR**TEACHER MOVES**

Consider prompting students to provide more information to justify their reasoning:

- Students can show their work and label each step of the work with a property.
- Prompt students to show their work and write a one sentence summary describing which property they used to justify equivalence.